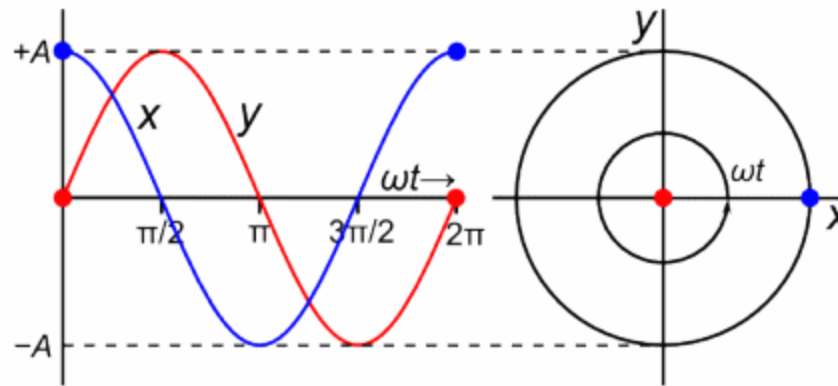




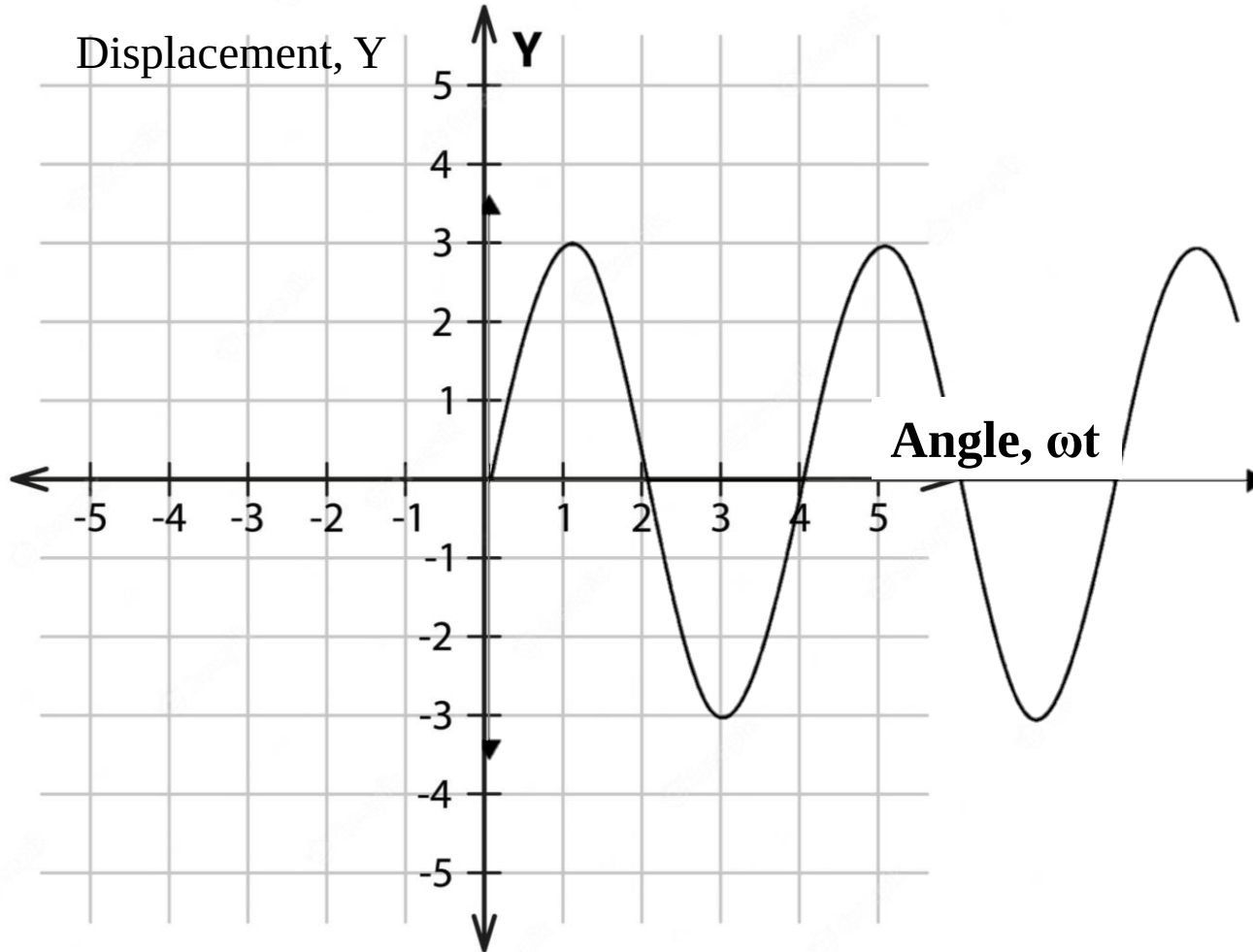
United International University

A Complete Guideline for Graphical Phase Change analysis in Simple Harmonic Motion

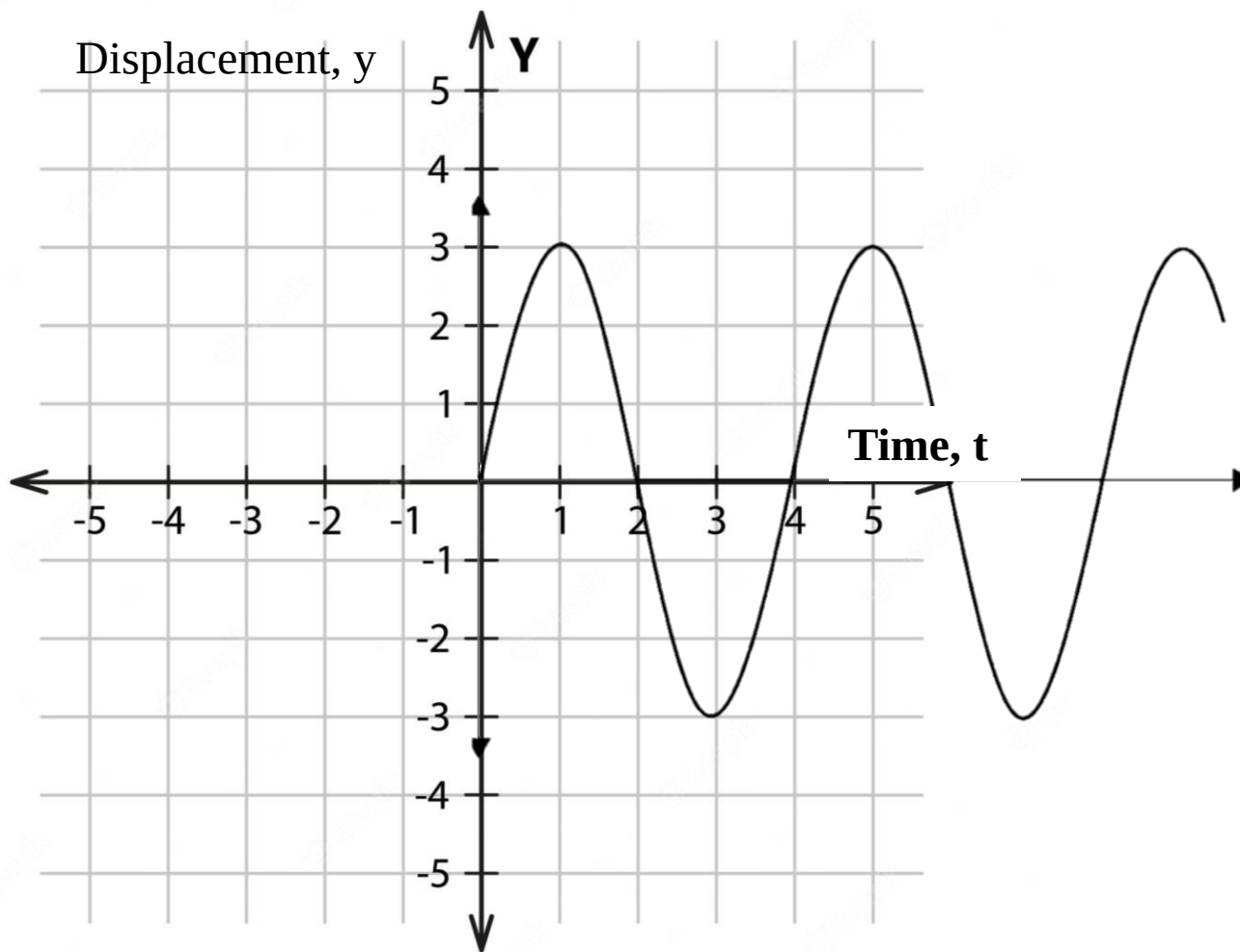


Let us consider the displacement of a simple harmonic oscillator is, $y = A \sin \omega t$

Let's see the graph can be represented as



We can also plot this $y = A \sin \omega t$ graph for y vs. t and the graph will be like the figure below



Now if we consider 2nd particle with $y = A \sin(\omega t + \phi)$

Displacement, $y = A \sin(\omega t + \phi)$

Now if we want to calculate the value of ωt for which the value of $y = 0$

For $y = 0$, $\sin(\omega t + \phi) = 0$

$$(\omega t + \phi) = 0$$

$$\omega t = -\phi \dots \dots \dots (1)$$

So the wave pattern of the particle will originate from an angle, $\omega t = -\phi$
if we consider four cases of displacements for graphical representation

(i) $y = A \sin(\omega t + \frac{\pi}{4})$

(ii) $y = A \sin(\omega t - \frac{\pi}{4})$

(iii) $y = A \sin(\omega t + \frac{\pi}{2})$

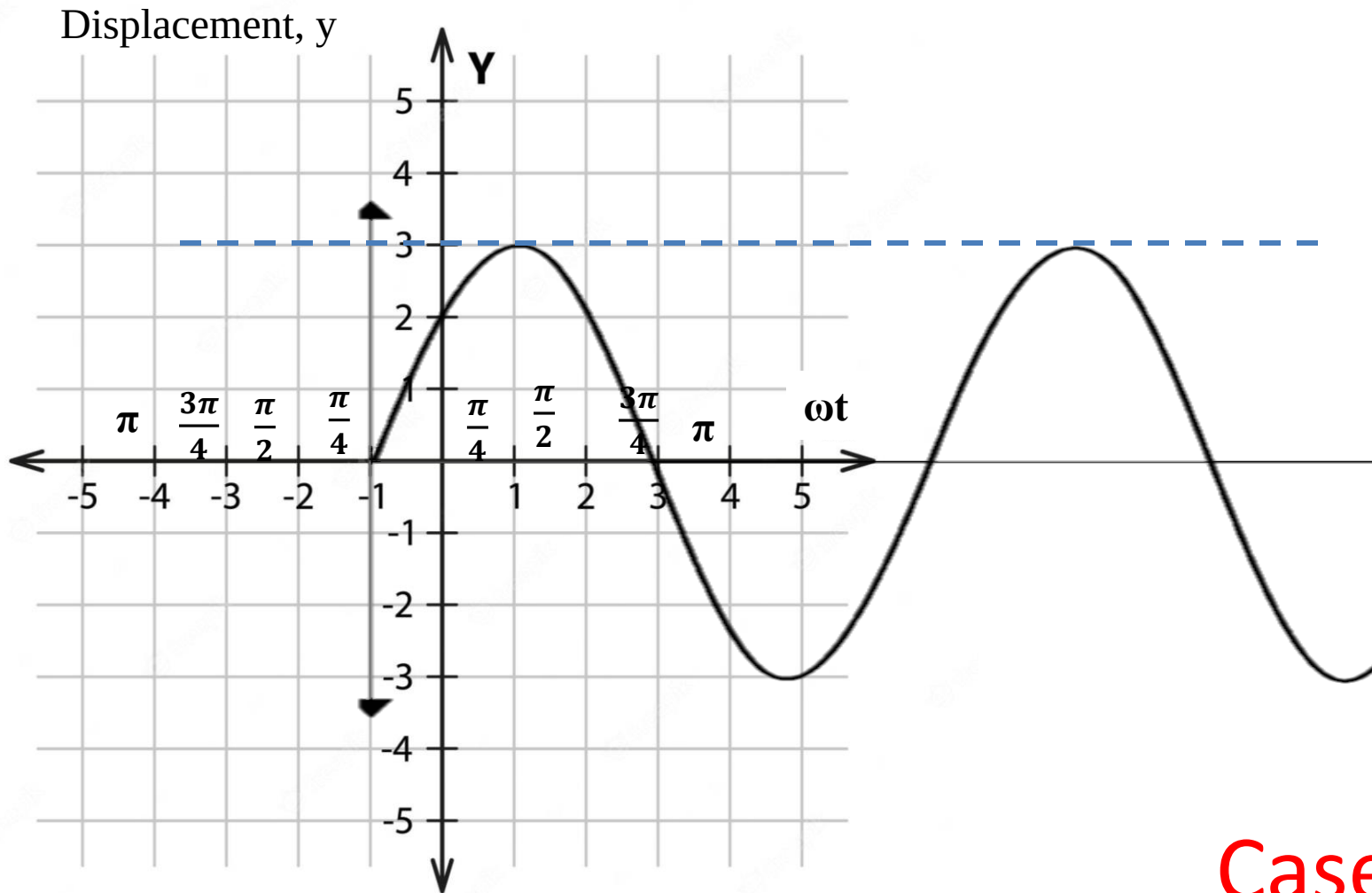
(iv) $y = A \sin(\omega t + \frac{\pi}{3})$

Please note that, you must plot the graph from $\omega t = 0$ position. We have only provided the negative value of ωt for primary understanding and practice purpose. But you must start the graph from $\omega t = 0$ when you understand the patterns.

Case (i)

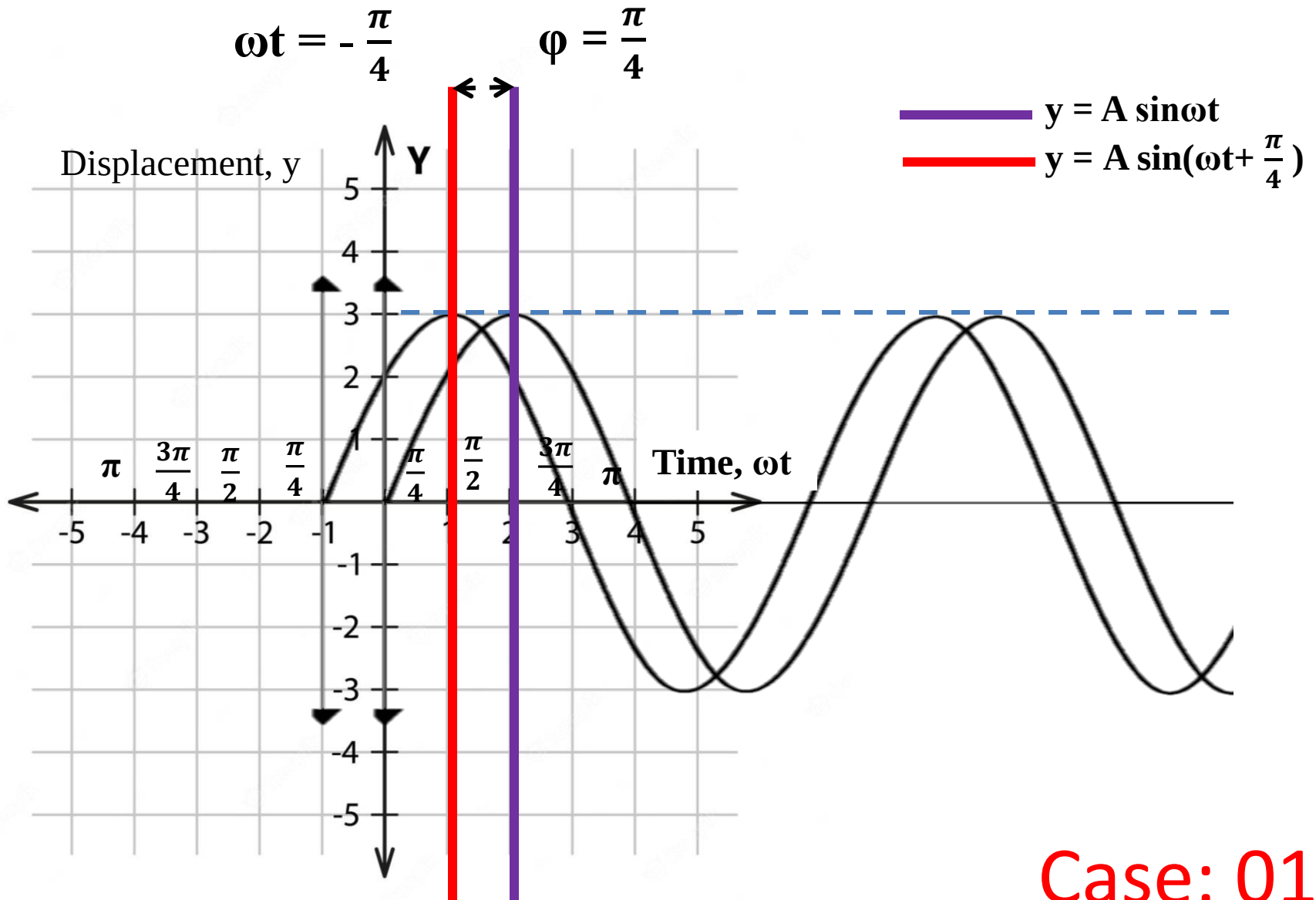
$$y = A \sin(\omega t + \frac{\pi}{4})$$

From Eqⁿ (1), the graph will start from $\omega t = -\frac{\pi}{4}$



Case: 01

Displacement y vs. Angle ωt

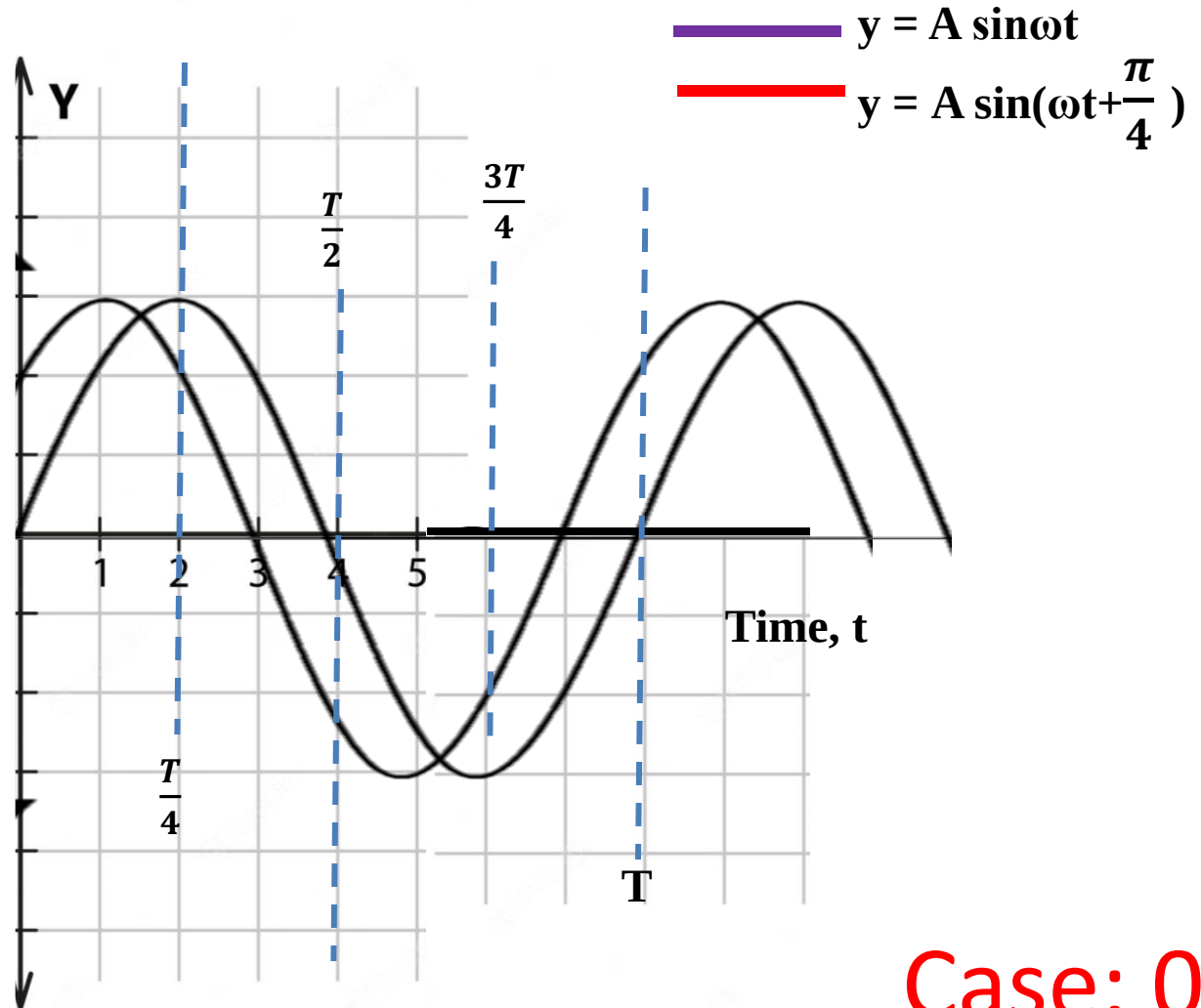


Case: 01

Displacement y vs. Time t

$$\omega t = -\frac{\pi}{4} \quad \phi = \frac{\pi}{4}$$

Displacement, y

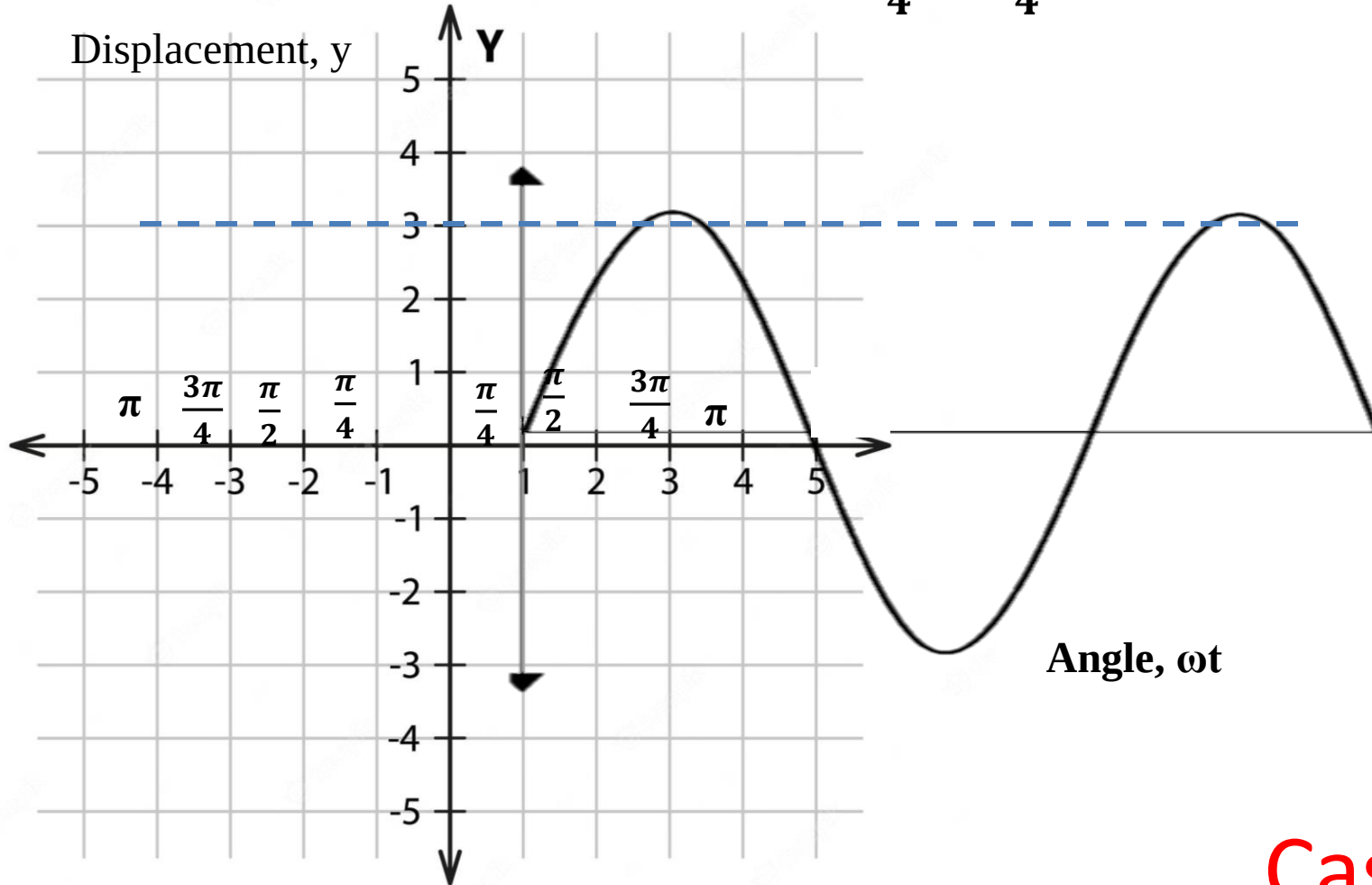


Case: 01

Case (ii)

$$y = A \sin(\omega t - \frac{\pi}{4})$$

From Eqⁿ (1), the graph will start from $\omega t = -(-\frac{\pi}{4}) = \frac{\pi}{4}$

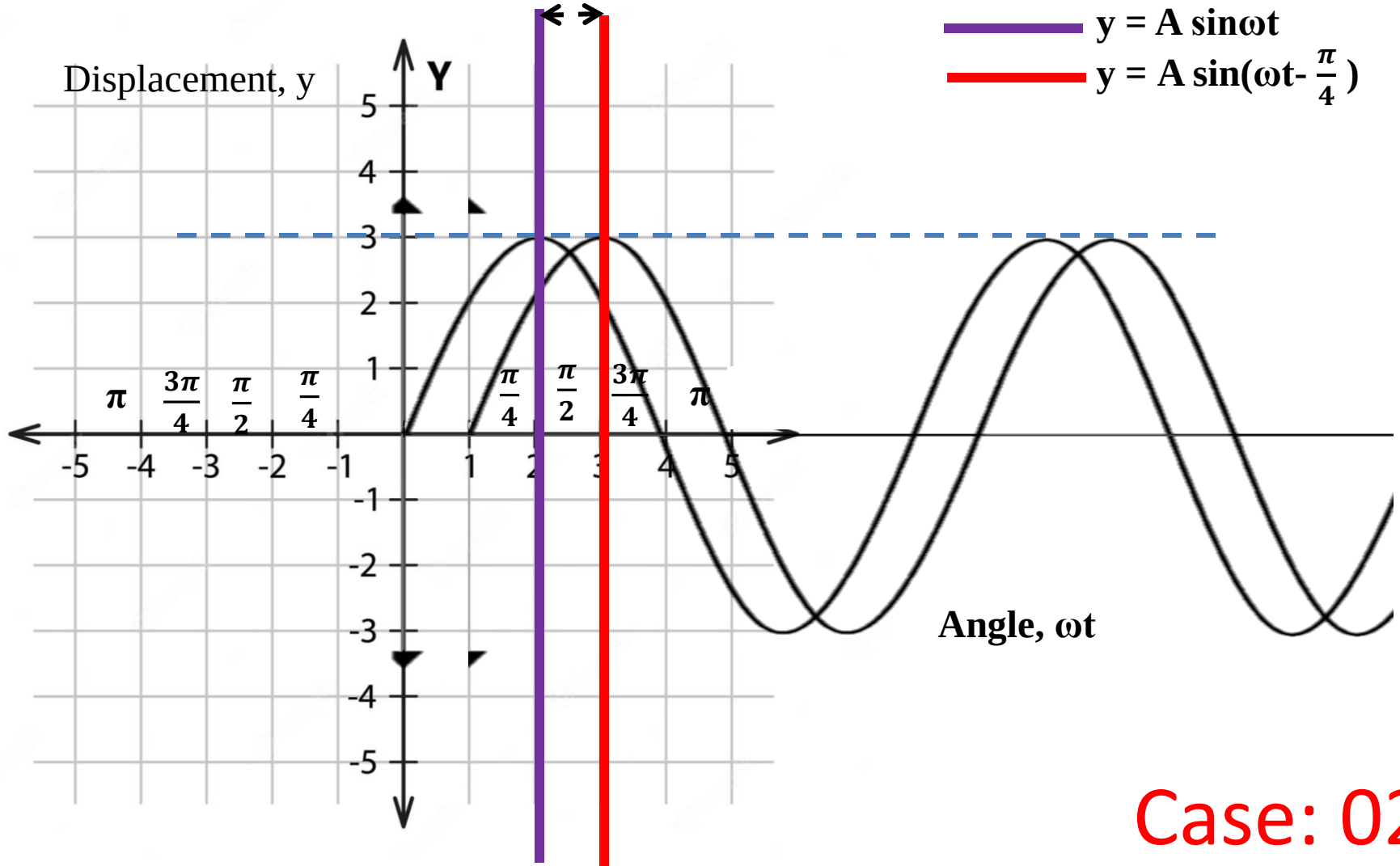


Case: 02

Displacement y vs. Angle ωt

$$\omega t = \frac{\pi}{4}$$

$$\phi = -\frac{\pi}{4}$$



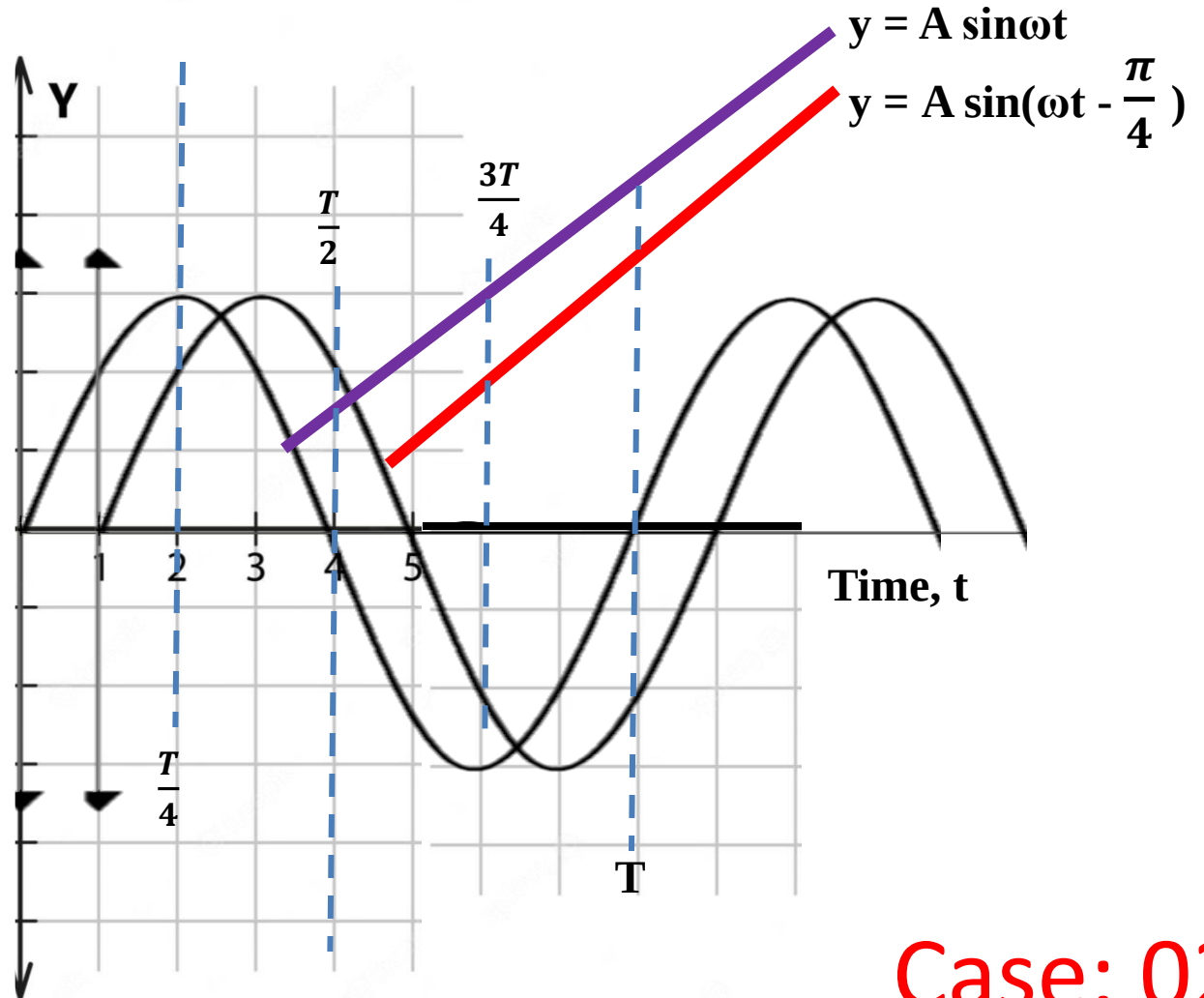
Case: 02

Displacement y vs. Time t

$$\omega t = \frac{\pi}{4}$$

$$\phi = -\frac{\pi}{4}$$

Displacement, y

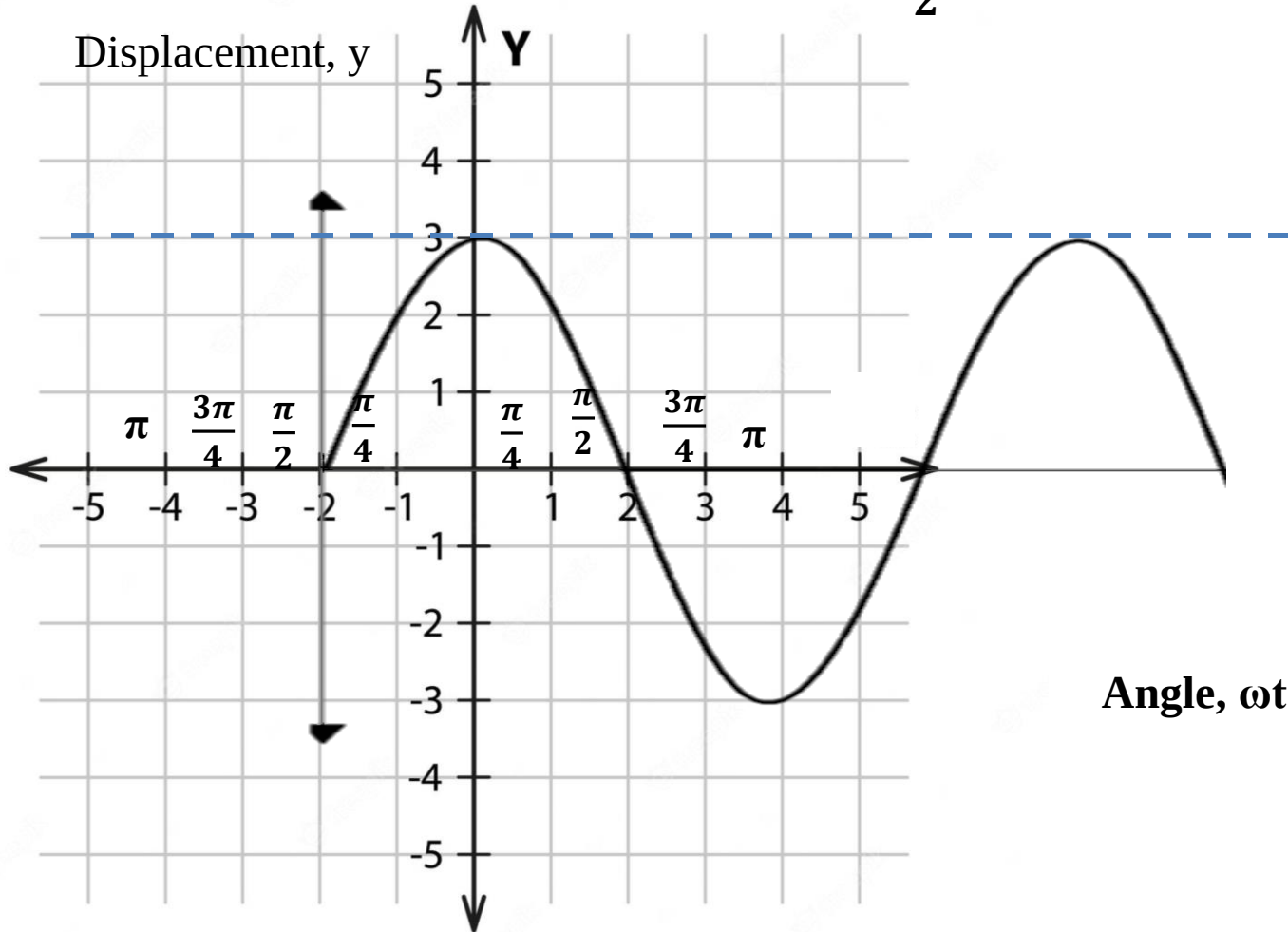


Case: 02

Case (iii)

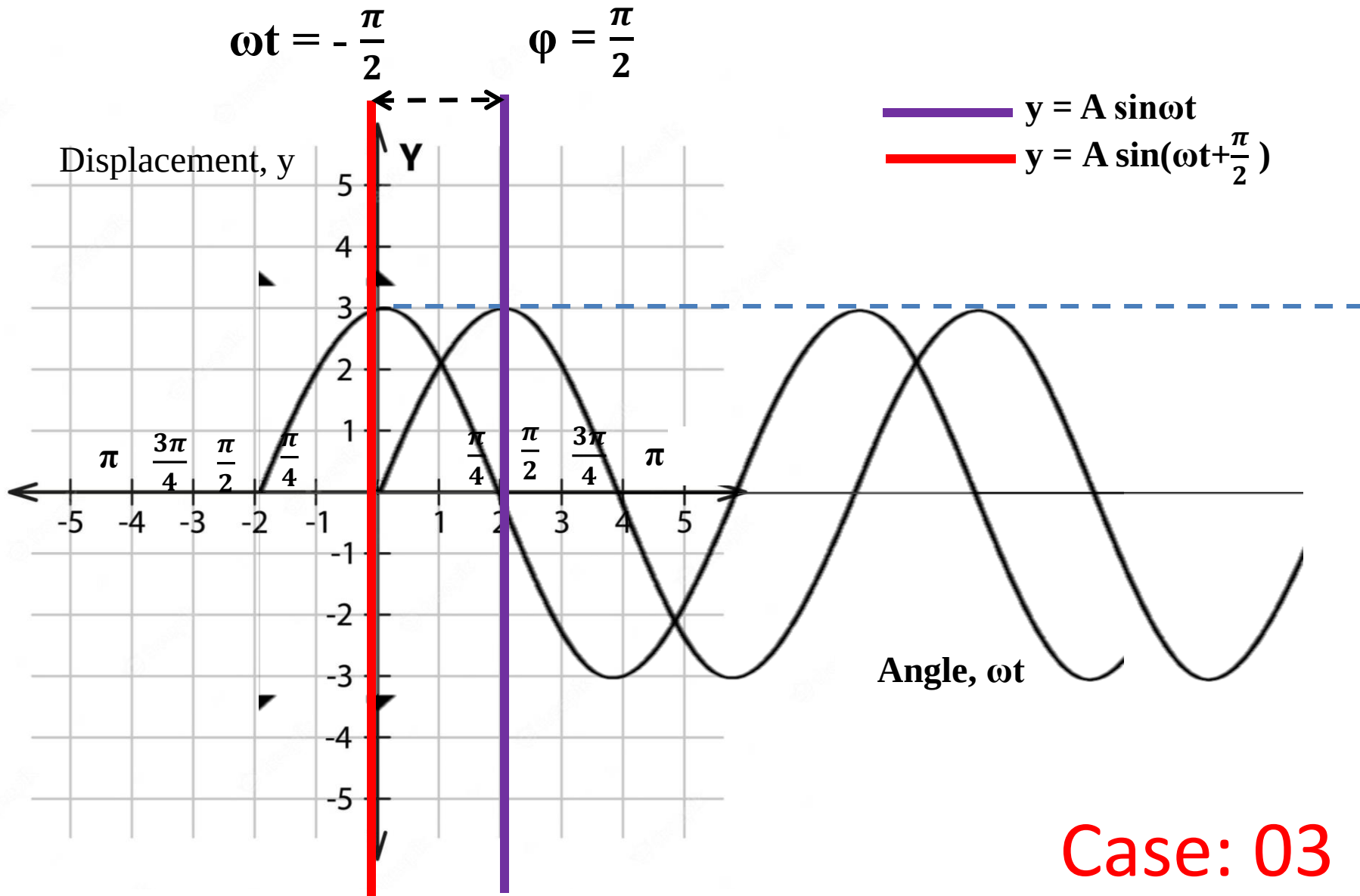
$$y = A \sin(\omega t + \frac{\pi}{2})$$

From Eqⁿ (1), the graph will start from $\omega t = -(\frac{\pi}{2})$



Case: 03

Displacement y vs. Angle ωt

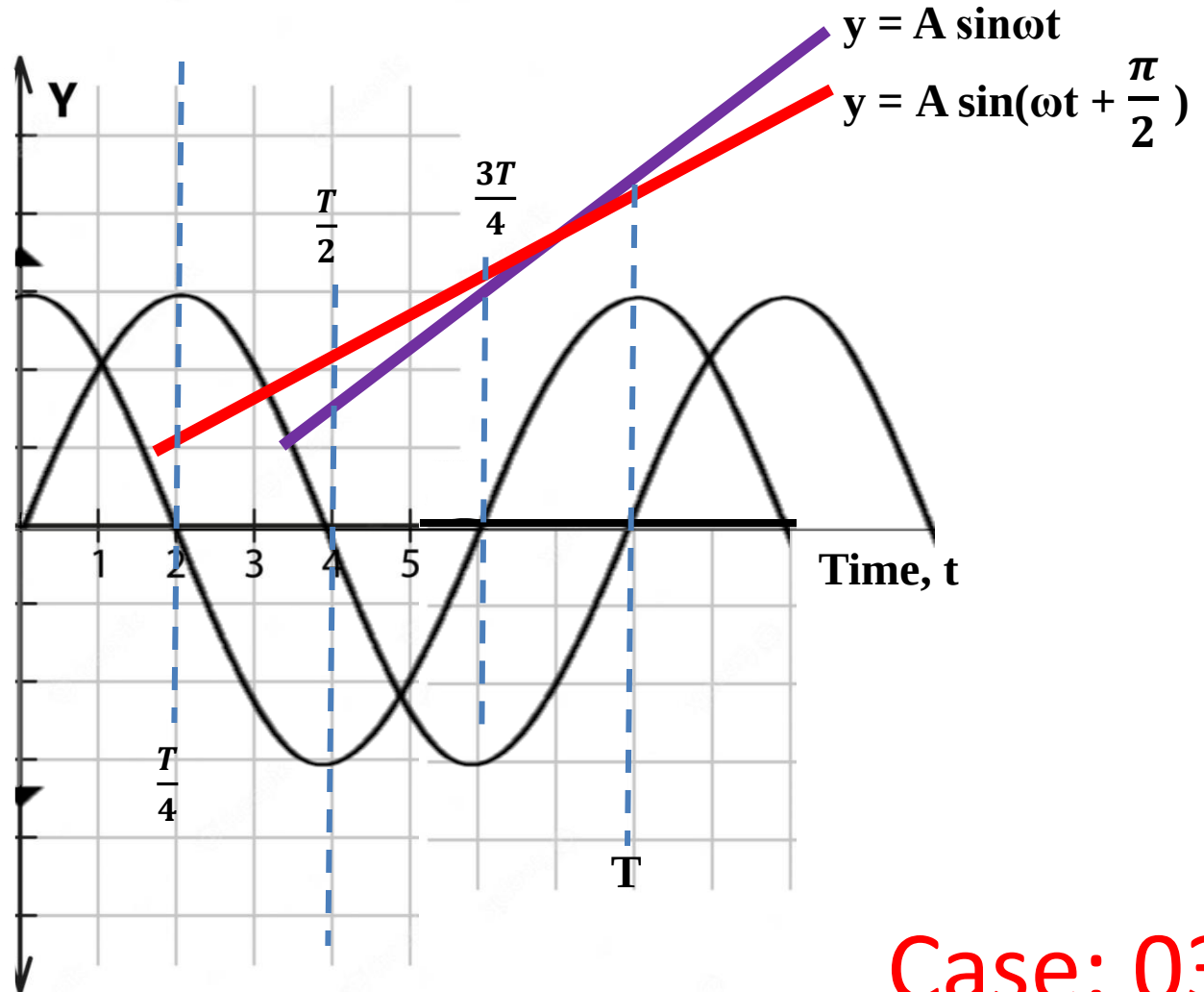


Case: 03

Displacement y vs. Time t

$$\omega t = -\frac{\pi}{2} \quad \phi = \frac{\pi}{2}$$

Displacement, y

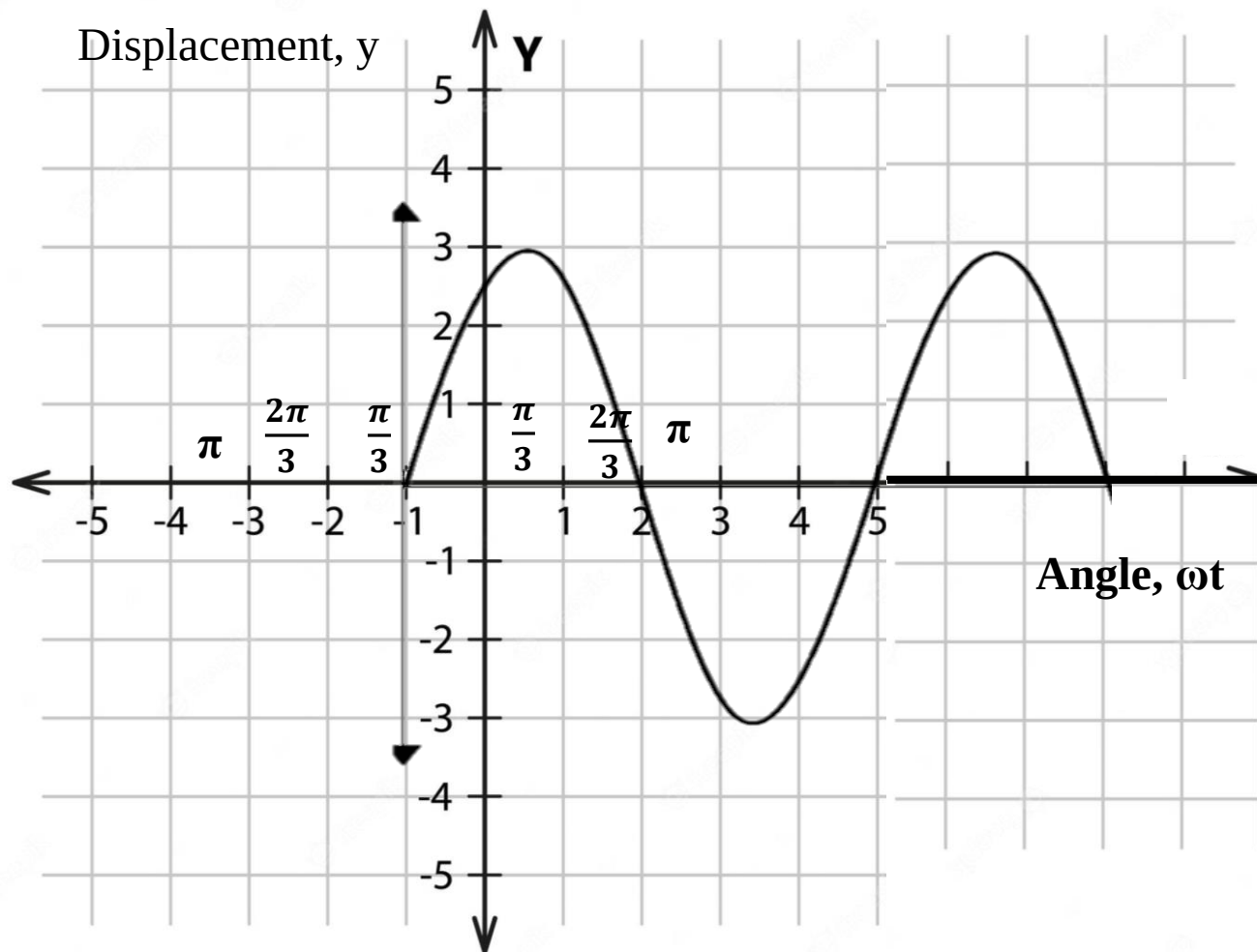


Case: 03

Case (iv):

Now if we consider the particle with $y = A \sin(\omega t + \frac{\pi}{3})$

graph for y vs. ωt and the graph will be like the figure below

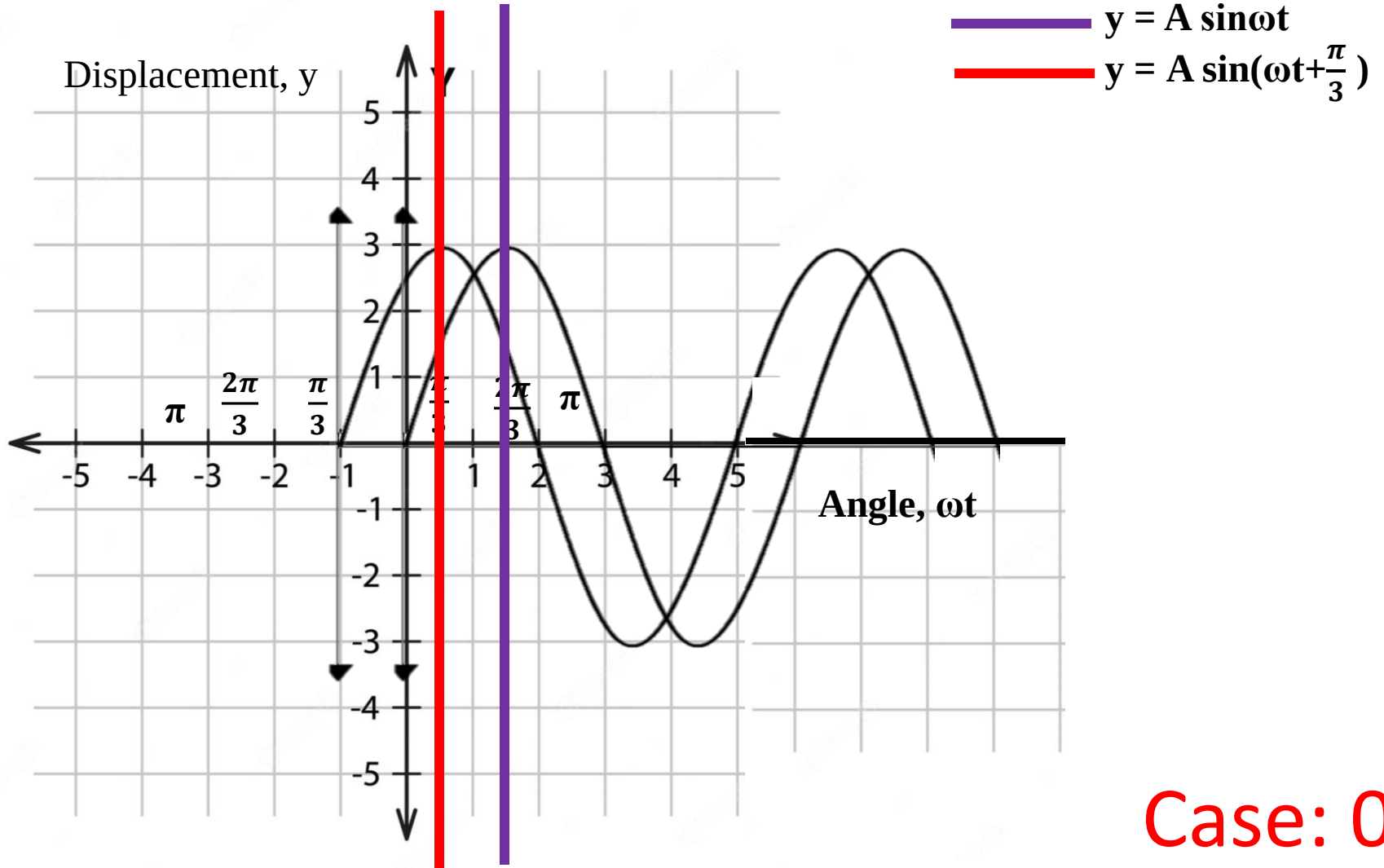


Case: 04

Displacement y vs. Angle ωt

$$\omega t = -\frac{\pi}{3}$$

$$\phi = \frac{\pi}{3}$$



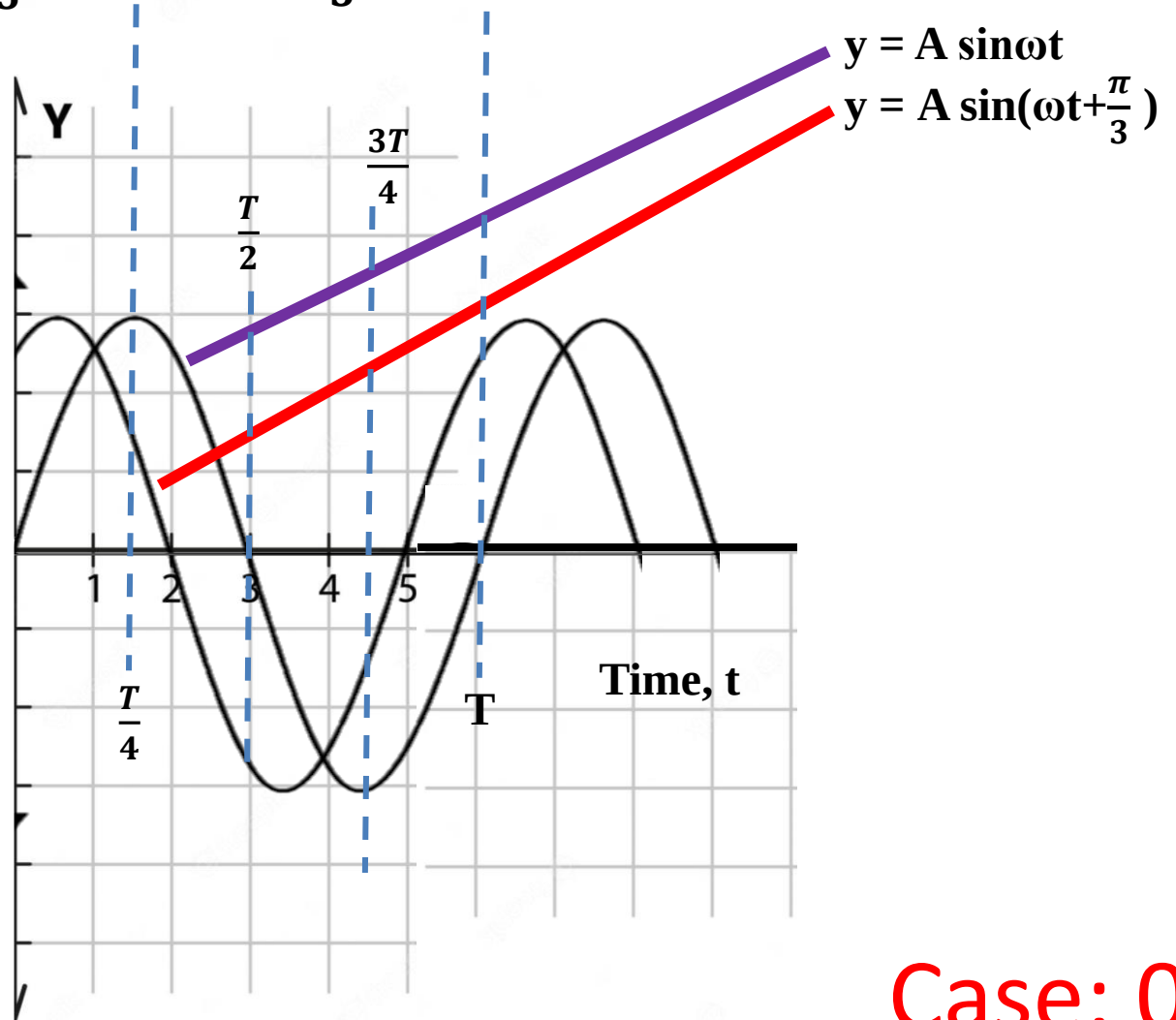
Case: 04

Displacement y vs. time t

$$\omega t = -\frac{\pi}{3}$$

$$\phi = \frac{\pi}{3}$$

Displacement, y



Case: 04

Some Practice Problems

(i) $y = A \sin(\omega t - \frac{\pi}{3})$

(ii) $y = A \sin(\omega t - \frac{\pi}{2})$

(iii) $y = A \sin(\omega t + \pi)$

(iv) $y = A \sin(\omega t - \pi)$

(v) $y = A \sin(\omega t + \frac{3\pi}{4})$

(vi) $y = A \sin(\omega t - \frac{3\pi}{4})$

(vii) $y = A \sin(\omega t + \frac{3\pi}{2})$

(viii) $y = A \sin(\omega t + 2\pi)$